CoKleisli Structure and Strength for Directed Container Comonads

Extracted from a machine-checked development

Abstract

We equip the comonad arising from a directed container with its coKleisli operations and its canonical strength. Given a directed container, the extension functor carries a comonad structure; we define coKleisli extension and composition, prove the unit and associativity laws of the coKleisli category, and construct the strength $\sigma : W(A \times B) \to A \times W(B)$ together with its naturality and counit laws. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

A directed container, in the sense of Ahman, Chapman, and Uustalu, is a container equipped with a root choice, a sub-shape operation, and a shift on positions, subject to five equational axioms (D1–D5). These data induce a comonad on the extension of the underlying container. The coKleisli category of that comonad is the natural home for "contextual" morphisms $W(X) \to Y$, and the coKleisli extension operation is the unique lift of such a morphism to an endomorphism of W.

This note isolates the coKleisli layer and the canonical strength of the comonad. We work throughout with a fixed directed container D, writing W for the extension of its underlying container, ε for the counit (extract at the root), and δ for the comultiplication (duplicate via sub-shapes and shifts). No special categorical machinery beyond the notions of comonad, coKleisli category, and strength is required. Proofs are indicated by the phrase "by lean," meaning they have been checked by a proof assistant rather than written out in prose.

What this note claims. In ordinary mathematical language: for any directed container, coKleisli extension satisfies the three standard laws

$$\operatorname{extend}\varepsilon = \mathrm{id}, \qquad \varepsilon \circ \operatorname{extend} f = f, \qquad \operatorname{extend}(g \circ \operatorname{extend} f) = \operatorname{extend} g \circ \operatorname{extend} f,$$

and coKleisli composition is therefore a unital associative operation on maps of the form $W(X) \to Y$. Moreover the strength

$$\sigma\langle s,\, v\rangle \;=\; \big(\pi_1\big(v(\mathrm{o}s)\big),\; \langle s,\, \pi_2 \circ v\rangle\big)$$

is natural in both variables and satisfies $(\mathrm{id} \times \varepsilon) \circ \sigma = \varepsilon$. All of this is formal content of the directed-container comonad; nothing further is required.

2 Background: directed containers and their comonads

We recall only the data used below. A *container* is a pair (S, P) consisting of a type of shapes and a family of position types. Its *extension* at a type X is

$$\mathrm{Ext}\, C\, X \;=\; \sum_{s\,:\,S} (P(s) \to X).$$

A *directed container* equips a container with operations

$$\begin{aligned}
\mathrm{o} &: (s : S) \to P(s),\\
\downarrow &: (s : S) \to P(s) \to S,\\
\oplus &: (s : S) \to (p : P(s)) \to P(s \downarrow p) \to P(s),
\end{aligned}$$

subject to five axioms D1–D5 (root-sub, root-shift, sub-root, nesting of subs, and associativity of shift). These induce a comonad on $\mathrm{Ext}\, C$:

$$\begin{aligned}
\varepsilon\langle s,\, v\rangle &\;=\; v(\mathrm{o}s),\\
\delta\langle s,\, v\rangle &\;=\; \langle s,\; \lambda p.\, \langle s \downarrow p,\; \lambda q.\, v(p \oplus q)\rangle\rangle.
\end{aligned}$$

We write W for this comonad (i.e. $W(X) = \mathrm{Ext}\, C\, X$) when the directed container is fixed.

3 CoKleisli extension

Definition 3.1 (CoKleisli extension)**.** Let D be a directed container and let $f : W(X) \to Y$ be a function. The **coKleisli extension** of f is the map

$$\mathrm{extend}\, f \;:\; W(X) \;\longrightarrow\; W(Y) \tag{1}$$

defined by

$$\mathrm{extend}\, f\, \langle s,\, v\rangle \;=\; \langle s,\; \lambda p.\, f\langle s \downarrow p,\; \lambda q.\, v(p \oplus q)\rangle\rangle. \tag{2}$$

Remark 3.2. Definitionally, $\mathrm{extend}\, f = W(f) \circ \delta$: one first duplicates the context and then applies f at every position. This is the standard coKleisli extension of a comonad.

Proposition 3.3 (Extension of the counit)**.** *For every type X,*

$$\mathrm{extend}\, \varepsilon \;=\; \mathrm{id}_{W(X)}. \tag{3}$$

Proof. By lean. □

Remark 3.4. Evaluating at $\langle s, v\rangle$ yields $\langle s,\, \lambda p.\, v\big(p \oplus \mathrm{o}(s \downarrow p)\big)\rangle$, which equals $\langle s, v\rangle$ by D3.

Proposition 3.5 (Counit after extension)**.** *For every $f : W(X) \to Y$,*

$$\varepsilon \circ \mathrm{extend}\, f \;=\; f. \tag{4}$$

Proof. By lean. □

Remark 3.6. This is the left unit law of the coKleisli category, and it characterises extend f as the unique map h with $\varepsilon \circ h = f$ compatible with the comultiplication in the usual way. The proof uses D1 and D2: extracting after extension probes at the root sub-shape, which is the original shape.

Proposition 3.7 (Extension respects composition)**.** *For $f : W(X) \to Y$ and $g : W(Y) \to Z$,*

$$\operatorname{extend}(g \circ \operatorname{extend} f) \;=\; \operatorname{extend} g \circ \operatorname{extend} f. \tag{5}$$

Proof. By lean. □

Remark 3.8. Both sides send $\langle s, v\rangle$ to a tree whose value at position p is g applied to a context built by nesting sub-shapes and shifts. Equality of shapes uses D4; equality of values uses D5. This is the associativity law of the coKleisli category, derived from coassociativity of δ.

4 CoKleisli composition

Definition 4.1 (CoKleisli composition)**.** Given coKleisli morphisms $f : W(X) \to Y$ and $g : W(Y) \to Z$, their **coKleisli composite** is

$$g \cdot_c f \;:=\; g \circ \operatorname{extend} f \;:\; W(X) \longrightarrow Z. \tag{6}$$

Proposition 4.2 (Right unit)**.** *For every $f : W(X) \to Y$,*

$$f \cdot_c \varepsilon \;=\; f. \tag{7}$$

Proof. By lean. □

Proposition 4.3 (Left unit)**.** *For every $f : W(X) \to Y$,*

$$\varepsilon \cdot_c f \;=\; f. \tag{8}$$

Proof. By lean. □

Proposition 4.4 (Associativity)**.** *For $f : W(X) \to Y$, $g : W(Y) \to Z$, and $h : W(Z) \to U$,*

$$(h \cdot_c g) \cdot_c f \;=\; h \cdot_c (g \cdot_c f). \tag{9}$$

Proof. By lean. □

Taken together, Propositions 4.2–4.4 say that coKleisli morphisms form a category with identities given by the components of ε.

5 Canonical strength

Every comonad on a cartesian category is canonically strong. For a directed container comonad the strength admits a direct combinatorial description.

Definition 5.1 (Strength)**.** The **canonical strength** of the directed container comonad is the family of maps

$$\sigma_{A,B} \;:\; W(A \times B) \longrightarrow A \times W(B) \tag{10}$$

defined by

$$\sigma_{A,B}\,\langle s,\, v\rangle \;=\; \Big(\pi_1\big(v(\mathrm{o}s)\big),\; \langle s,\; \pi_2 \circ v\rangle\Big). \tag{11}$$

The first component extracts the A-part at the root; the second retains the shape and projects every label to its B-part.

Proposition 5.2 (Naturality in the left variable)**.** *For every $f : A \to A'$,*

$$(f \times \mathrm{id}) \circ \sigma_{A,B} \;=\; \sigma_{A',B} \circ W(f \times \mathrm{id}). \tag{12}$$

Proof. By lean. □

Proposition 5.3 (Naturality in the right variable)**.** *For every $g : B \to B'$,*

$$(\mathrm{id} \times W(g)) \circ \sigma_{A,B} \;=\; \sigma_{A,B'} \circ W(\mathrm{id} \times g). \tag{13}$$

Proof. By lean. □

Proposition 5.4 (Strength and counit)**.**

$$(\mathrm{id} \times \varepsilon) \circ \sigma_{A,B} \;=\; \varepsilon_{A \times B}. \tag{14}$$

Proof. By lean. □

Remark 5.5. Extracting the B-component of the strength is the same as extracting directly from the $A \times B$-labelled tree: both pick the B-part of the label at the root.

6 Discussion

What has been established. The coKleisli category of a directed container comonad has been constructed explicitly from root, sub, and shift. The unit laws rest on D1–D3; the associativity law rests on D4–D5. The strength is the standard one for a comonad on a cartesian category, written out in container coordinates.

Scope. The development treats a fixed directed container and does not re-derive the comonad laws themselves (those belong to the underlying directed-container theory). Strength is stated only with the counit interaction recorded above; further strength axioms (e.g. interaction with comultiplication) are not listed separately here.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant; we have not reproduced the formal proof scripts here.

Acknowledgements

The mathematical content follows the theory of directed containers of Ahman, Chapman, and Uustalu, and standard coKleisli calculus for comonads. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

[1] D. Ahman, J. Chapman, and T. Uustalu. When is a container a comonad? *Logical Methods in Computer Science*, 10(3), 2014.

[2] M. Abbott, T. Altenkirch, and N. Ghani. Containers: constructing strictly positive types. *Theoretical Computer Science*, 342(1):3–27, 2005.